

## APALACHICOLA REGIONAL, FL, USA

WMO: 722200

Lat: 29.733N

Lon: 85.033W

Elev: 6

StdP: 101.26

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 12832

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -0.2       | 1.8 | -8.7                          | 1.8 | 4.0  | -6.1 | 2.2 | 5.8  | 9.9                   | 14.1 | 9.0 | 13.9 | 2.6                   | 50   | 0.401 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.6                    | 33.5            | 26.0 | 32.7 | 25.9 | 32.1 | 25.8 | 27.7                | 31.2 | 27.2 | 30.8 | 26.8 | 30.3 | 4.8                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.7                            | 22.3 | 29.3 | 26.2 | 21.7 | 29.1 | 25.8 | 21.1 | 28.9 | 87.4          | 31.5 | 85.5 | 30.9 | 83.6 | 30.2 | 29.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 8.6               | 7.7  | 7.0 | -3.3                       | 35.9 | 1.7                | 2.5 | -4.5                                               | 37.7 | -5.5       | 39.1 | -6.4       | 40.5 | -7.6       | 42.3 |    |    |
|                   |      |     | -5.0                       | 28.5 | 1.7                | 0.8 | -6.2                                               | 29.0 | -7.1       | 29.5 | -8.1       | 29.9 | -9.3       | 30.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days and<br>Degree-Hours                              | DBAvg       | 20.7   | 12.1  | 13.8  | 16.5  | 19.8  | 23.9  | 27.1  | 27.8  | 27.9  | 26.3  | 22.1  | 16.7  | 13.8 | (6)  |
|                                                                               | DBStd       | 6.63   | 4.91  | 4.32  | 3.87  | 3.17  | 2.48  | 1.58  | 1.33  | 1.35  | 1.92  | 3.71  | 4.12  | 4.89 | (7)  |
|                                                                               | HDD10.0     | 74     | 36    | 13    | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 19   | (8)  |
|                                                                               | HDD18.3     | 677    | 197   | 133   | 80    | 21    | 1     | 0     | 0     | 0     | 0     | 13    | 78    | 153  | (9)  |
|                                                                               | CDD10.0     | 3974   | 102   | 121   | 205   | 293   | 431   | 512   | 553   | 556   | 488   | 374   | 203   | 138  | (10) |
|                                                                               | CDD18.3     | 1535   | 4     | 7     | 24    | 64    | 173   | 262   | 294   | 297   | 238   | 128   | 28    | 14   | (11) |
|                                                                               | CDH23.3     | 14782  | 2     | 11    | 56    | 323   | 1551  | 2819  | 3314  | 3302  | 2261  | 998   | 121   | 23   | (12) |
|                                                                               | CDH26.7     | 5163   | 0     | 0     | 3     | 25    | 404   | 1081  | 1343  | 1312  | 772   | 213   | 7     | 4    | (13) |
| Wind                                                                          | WSAvg       | 3.2    | 3.4   | 3.4   | 3.6   | 3.6   | 3.5   | 3.2   | 2.7   | 2.8   | 3.1   | 3.2   | 3.2   | 3.4  | (14) |
| Precipitation                                                                 | PrecAvg     | 1389   | 106   | 98    | 117   | 66    | 64    | 128   | 176   | 191   | 180   | 98    | 82    | 88   | (15) |
|                                                                               | PrecMax     | 2242   | 528   | 251   | 329   | 308   | 308   | 416   | 459   | 479   | 477   | 521   | 191   | 246  | (16) |
|                                                                               | PrecMin     | 824    | 19    | 11    | 0     | 1     | 0     | 5     | 19    | 59    | 14    | 1     | 5     | 21   | (17) |
|                                                                               | PrecStd     | 335    | 83    | 59    | 77    | 70    | 59    | 100   | 110   | 100   | 120   | 99    | 50    | 50   | (18) |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 23.2  | 24.9  | 26.7  | 28.6  | 32.6  | 34.7  | 34.7  | 34.5  | 33.0  | 31.1  | 27.5  | 25.2 | (19) |
|                                                                               |             | MCWB   | 20.3  | 20.0  | 19.7  | 21.6  | 23.7  | 26.1  | 26.3  | 26.3  | 25.3  | 24.8  | 22.8  | 21.4 | (20) |
|                                                                               | 2%          | DB     | 22.0  | 22.9  | 24.8  | 27.4  | 31.1  | 33.0  | 33.1  | 33.2  | 32.0  | 30.0  | 26.1  | 23.7 | (21) |
|                                                                               |             | MCWB   | 19.1  | 19.4  | 19.9  | 21.6  | 23.7  | 25.8  | 26.2  | 26.2  | 25.1  | 24.3  | 22.0  | 20.9 | (22) |
|                                                                               | 5%          | DB     | 20.9  | 21.6  | 23.6  | 26.3  | 29.9  | 32.2  | 32.5  | 32.4  | 31.3  | 28.9  | 24.8  | 22.5 | (23) |
|                                                                               |             | MCWB   | 18.3  | 18.7  | 19.4  | 20.9  | 23.4  | 25.6  | 26.1  | 26.2  | 25.0  | 23.7  | 20.8  | 20.4 | (24) |
|                                                                               | 10%         | DB     | 19.4  | 20.4  | 22.5  | 25.2  | 28.8  | 31.2  | 31.7  | 31.7  | 30.4  | 27.8  | 23.6  | 21.4 | (25) |
|                                                                               |             | MCWB   | 17.3  | 17.9  | 18.9  | 20.7  | 23.1  | 25.2  | 26.0  | 26.1  | 24.8  | 23.2  | 20.2  | 19.7 | (26) |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 21.4  | 21.8  | 22.7  | 24.7  | 26.1  | 27.9  | 28.3  | 28.2  | 27.8  | 27.2  | 24.6  | 22.9 | (27) |
|                                                                               |             | MCDB   | 22.4  | 23.1  | 24.4  | 26.7  | 28.8  | 31.0  | 31.4  | 31.8  | 30.7  | 28.8  | 26.4  | 23.8 | (28) |
|                                                                               | 2%          | WB     | 20.1  | 20.7  | 21.6  | 23.8  | 25.3  | 27.1  | 27.5  | 27.6  | 26.9  | 25.8  | 23.3  | 22.0 | (29) |
|                                                                               |             | MCDB   | 21.3  | 22.0  | 23.3  | 25.8  | 28.7  | 30.7  | 31.5  | 31.3  | 29.9  | 28.2  | 25.0  | 23.0 | (30) |
|                                                                               | 5%          | WB     | 19.0  | 19.6  | 20.7  | 22.8  | 24.8  | 26.6  | 27.0  | 27.1  | 26.2  | 25.1  | 22.2  | 21.1 | (31) |
|                                                                               |             | MCDB   | 20.1  | 21.1  | 22.5  | 25.2  | 28.1  | 30.4  | 30.8  | 30.8  | 29.4  | 27.7  | 23.9  | 22.2 | (32) |
|                                                                               | 10%         | WB     | 17.9  | 18.6  | 19.7  | 21.9  | 24.2  | 26.0  | 26.6  | 26.7  | 25.7  | 24.3  | 21.1  | 19.8 | (33) |
|                                                                               |             | MCDB   | 19.0  | 19.7  | 21.6  | 24.0  | 27.6  | 29.8  | 30.3  | 30.3  | 28.9  | 27.1  | 23.0  | 21.2 | (34) |
| Mean Daily<br>Temperature<br>Range                                            |             | MDBR   | 9.9   | 9.7   | 9.6   | 9.6   | 9.2   | 7.9   | 7.6   | 7.3   | 7.9   | 9.7   | 10.7  | 9.7  | (35) |
|                                                                               | 5% DB       | MCDBR  | 8.8   | 9.1   | 9.2   | 9.5   | 10.0  | 9.1   | 8.6   | 8.3   | 8.7   | 9.1   | 9.9   | 8.4  | (36) |
|                                                                               |             | MCWBR  | 6.6   | 6.7   | 5.7   | 5.2   | 4.5   | 3.8   | 3.6   | 3.5   | 3.7   | 4.7   | 6.4   | 6.4  | (37) |
|                                                                               | 5% WB       | MCDBR  | 7.2   | 7.9   | 7.4   | 7.2   | 7.8   | 7.8   | 7.7   | 7.2   | 7.1   | 7.3   | 7.7   | 7.2  | (38) |
|                                                                               |             | MCWBR  | 6.0   | 6.6   | 5.3   | 5.1   | 4.1   | 3.7   | 3.6   | 3.5   | 3.5   | 4.3   | 5.9   | 6.2  | (39) |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.347  | 0.356 | 0.367 | 0.394 | 0.436 | 0.469 | 0.491 | 0.482 | 0.435 | 0.387 | 0.356 | 0.346 | (40) |      |
|                                                                               | taud        | 2.507  | 2.495 | 2.465 | 2.381 | 2.297 | 2.267 | 2.221 | 2.251 | 2.362 | 2.474 | 2.494 | 2.532 | (41) |      |
|                                                                               | Ebn at Noon | 887    | 910   | 920   | 901   | 859   | 827   | 807   | 811   | 842   | 865   | 868   | 871   | (42) |      |
|                                                                               | Edh at Noon | 90     | 99    | 108   | 122   | 133   | 137   | 143   | 137   | 119   | 100   | 90    | 84    | (43) |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 3.18   | 3.85  | 5.13  | 6.15  | 6.82  | 6.26  | 5.87  | 5.48  | 5.07  | 4.56  | 3.58  | 2.85  | (44) |      |
|                                                                               | RadStd      | 0.25   | 0.46  | 0.43  | 0.51  | 0.41  | 0.43  | 0.42  | 0.39  | 0.36  | 0.33  | 0.26  | 0.26  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | +0.60 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (26 neighbors) | +0.46 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -97     | +162    | +100    |

Nomenclature: See separate page